Feature demo · issue #167 · MCP

An agent saves a diagram — the picture is exact

Recorded with a real MCP client (stdio, as VS Code would run it), the real server, the real draw.io editor and the desktop app's own Chromium

- **The agent works on the file.** docs_read(diagram) writes the .drawio file, the agent edits it with its own tools, docs_write(diagram_save) stores it. Every draw.io feature is available: AWS icons, swimlanes, notes, dashed and routed connectors.
- **The picture comes from draw.io itself.** With no editor open, the server asks it for the SVG in a hidden browser: the desktop app's own executable (`--export-svg`, Electron's Chromium, nothing to install), or a Chrome / Chromium / Edge on the machine.
- **No browser?** A plain preview is drawn, the footer says "Simple preview", and the agent is told. The editor replaces it on the next save.
- **Safe by default.** The hidden page may only load the editor's own origin; the public diagrams.net is never used unless an admin switches it on.

1. The agent's session (real MCP calls)

Project and ticket created, then a 43-element order flow (4 swimlanes, AWS icons, decisions, a cylinder, a note, 19 routed connectors) saved from a file. The answer says how the picture was made.

Create the project

```
→ manage_project(action=create, name=Kanban Board, prefix=KAN, color=#2E6F40)
```


```
← {"id":"2e333609-abff-4ce2-ba4e-6f8bb81d057e","name":"Kanban Board","prefix":"KAN","color":"#2E6F40","ticket_counter":0,"repo_path":null,"worktree_template":nul
l,"worktree_by_default":false,"docs_enabled":true}
```


Create the ticket

```
→ manage_ticket(action=create, project_id=2e333609-abff-4ce2-ba4e-6f8bb81d057e, title=Document the order flow, type=task, priority=high, description=Order flow as t
he team agreed it. The diagram below was draw …)
```


```
← {"id":"KAN-1","project_id":"2e333609-abff-4ce2-ba4e-6f8bb81d057e","title":"Document the order flow","description":"Order flow as the team agreed it. The diagram b
elow was drawn by the agent.","type":"task","status":"backlog","prio …
```


Create the diagram from a .drawio file

```
→ docs_write(action=diagram_create, project_id=2e333609-abff-4ce2-ba4e-6f8bb81d057e, title=Order flow, path=/tmp/claude-0/demo2/work/order-flow.drawio, summary=Cust
omer to confirmation across edge, services and data)
```


```
← {"id":"dg_42f175c4","label":"Order flow","version":1,"markdown":"[Order flow](diagram:dg_42f175c4)","preview":"exact: drawn by the draw.io editor"}
```


Put the diagram in the ticket

```
→ manage_ticket(action=update, ticket_id=KAN-1, description=Order flow as the team agreed it. The diagram below was draw …)
```


```
← {"id":"KAN-1","project_id":"2e333609-abff-4ce2-ba4e-6f8bb81d057e","title":"Document the order flow","description":"Order flow as the team agreed it. The diagram b
elow was drawn by the agent.\n\n[Order flow](diagram:dg_42f175c4)\n" …
```


Export the diagram to a file to edit it

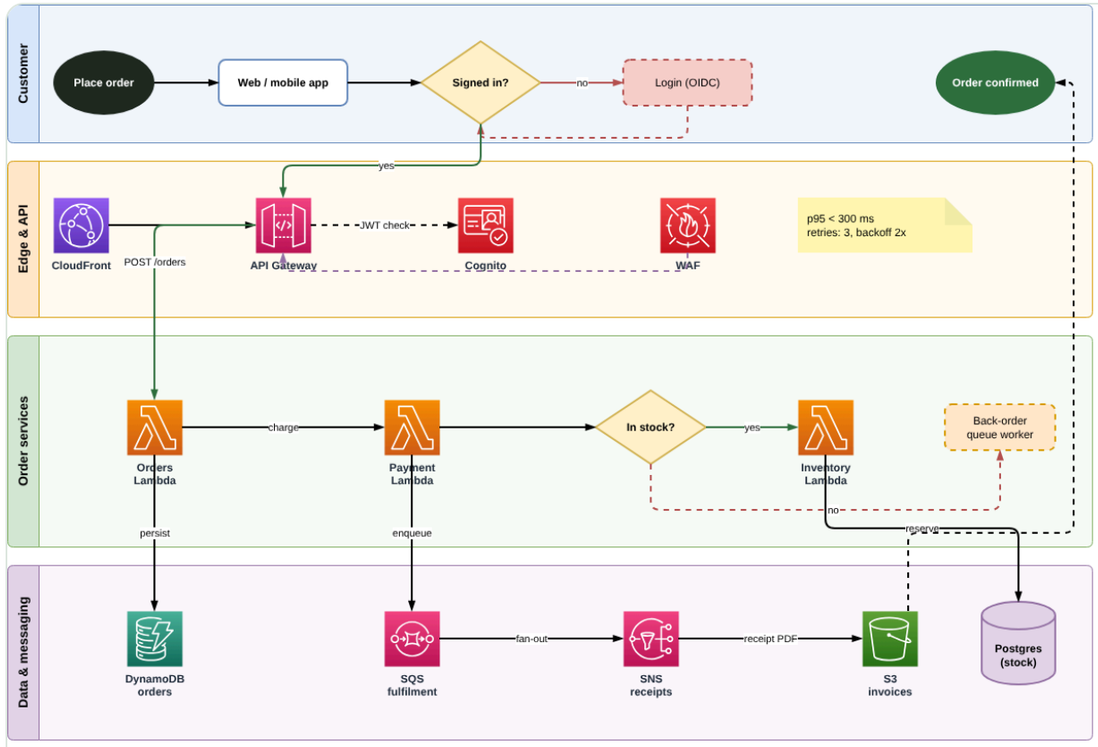
```
→ docs_read(action=diagram, diagram_id=dg_42f175c4, path=/tmp/claude-0/demo2/work/edit/)
```


```
← {"path":"/tmp/claude-0/demo2/work/edit/Order-flow.drawio","id":"dg_42f175c4","label":"Order flow","version":1,"markdown":"[Order flow](diagram:dg_42f175c4)"}
```

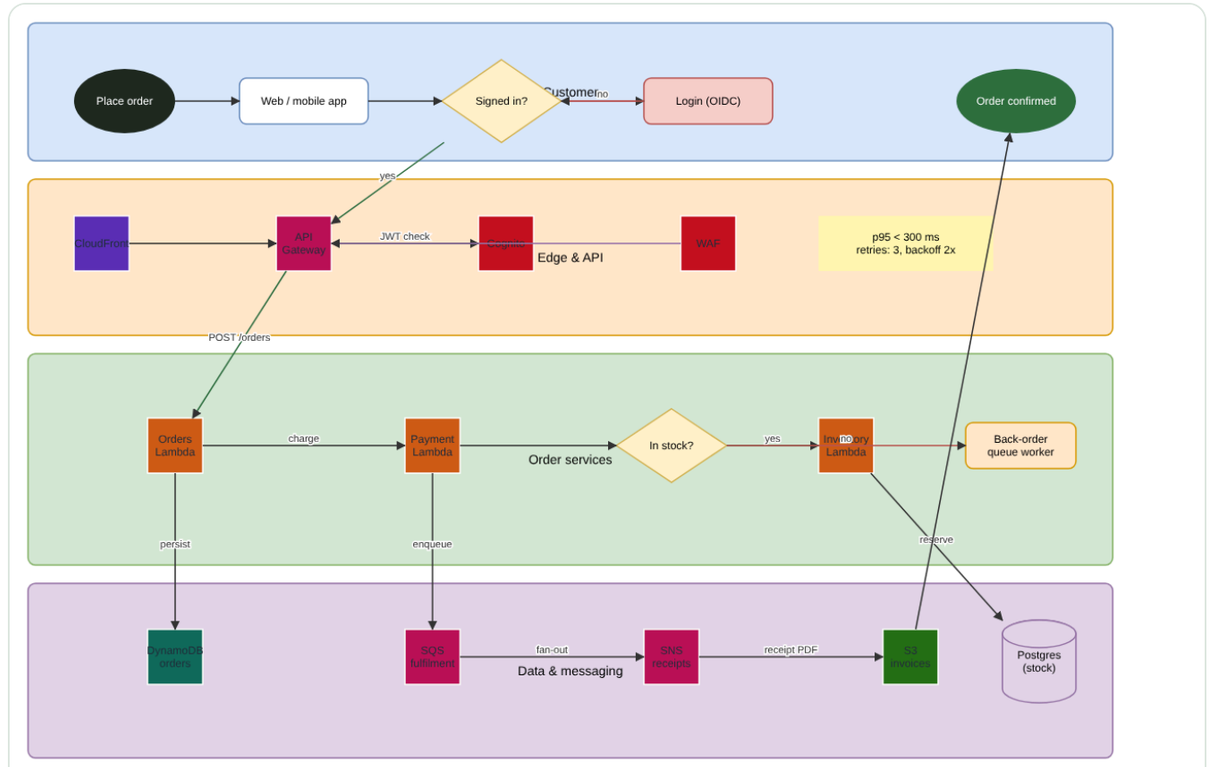
2. The same file, two pictures

Left: what the editor draws (what is stored when the desktop app or a browser is there). Right: the plain preview the server can draw alone: icons and containers become boxes.

Exact · drawn by draw.io

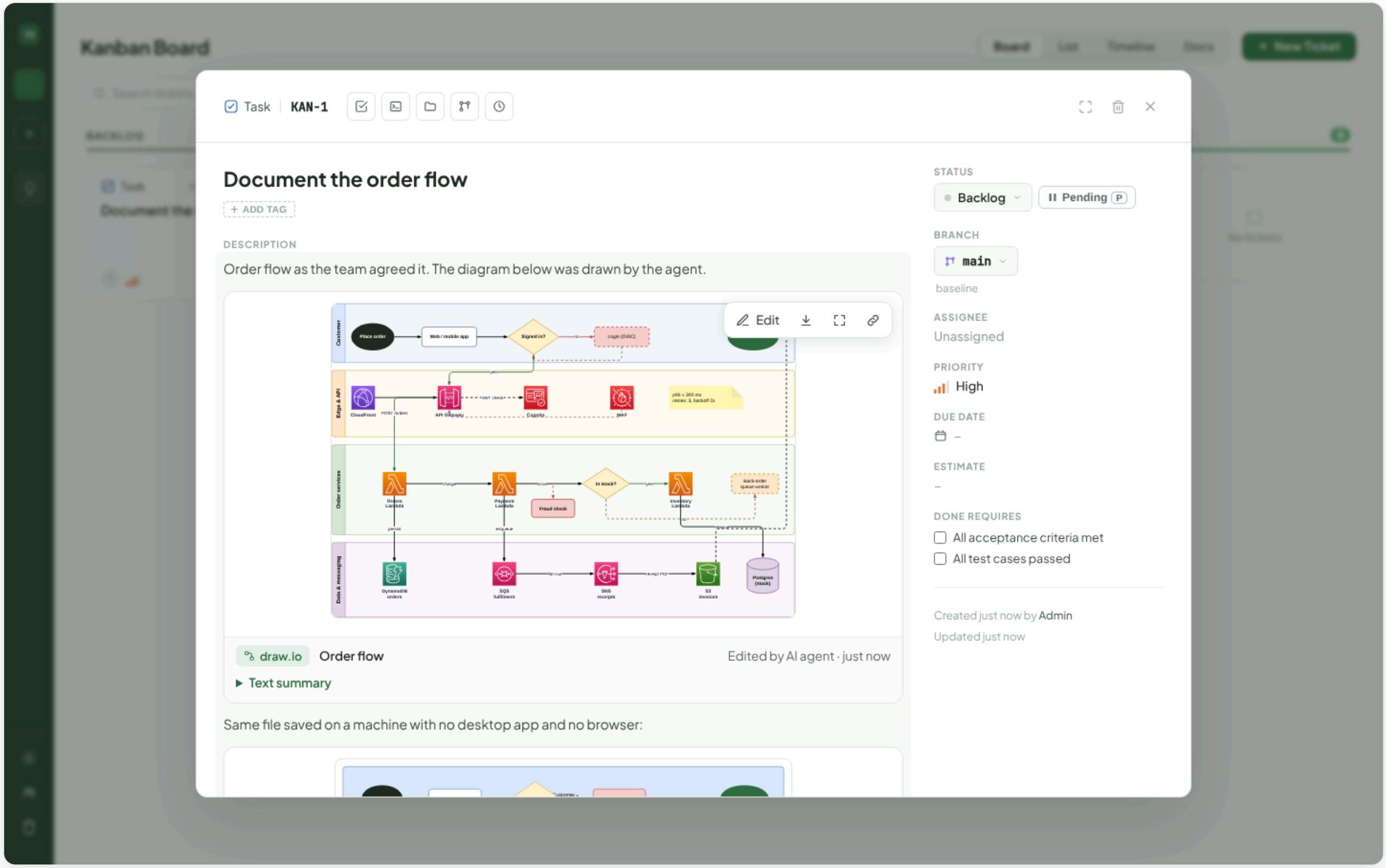


Simple preview · drawn by the server



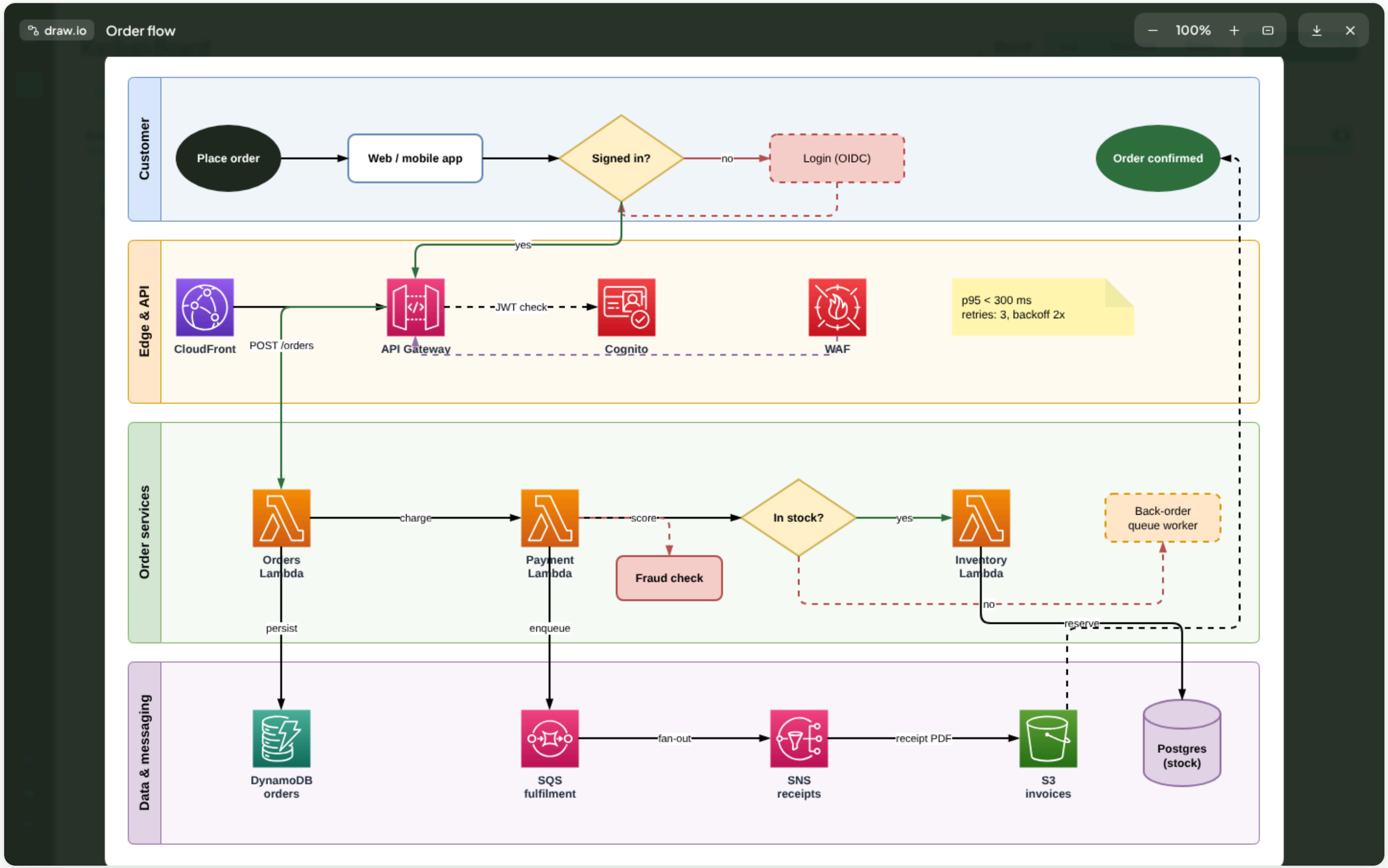
3. In the ticket, as stored by the agent

The agent put the returned `[Order flow](diagram:dg_…)` line in the description. The footer shows who saved it. (This is version 2: see step 5.)



4. Full size: every icon, lane and connector

Zoom and pan in the viewer. This is the stored SVG, made by the editor with no window open.



5. The agent edits the file and saves again

It exports the diagram, adds a "Fraud check" step with a connector (a file edit), tries a stale `base_version` (refused with what to do), then saves version 2. The picture is exact again.

Save with a stale base_version (rejected)

```
→ docs_write(action=diagram_save, diagram_id=dg_42f175c4, path=/tmp/claude-0/demo2/work/edit/Order-flow.drawio, base_version=0)
```

```
← Error executing tool docs_write: agent saved a newer version while you were editing. Export the diagram again, re-apply your change on the new version, and save i
t.
```

Save the edited file

```
→ docs_write(action=diagram_save, diagram_id=dg_42f175c4, path=/tmp/claude-0/demo2/work/edit/Order-flow.drawio, base_version=1)
```


```
← {"id":"dg_42f175c4","label":"Order flow","version":2,"preview":"exact: drawn by the draw.io editor"}
```

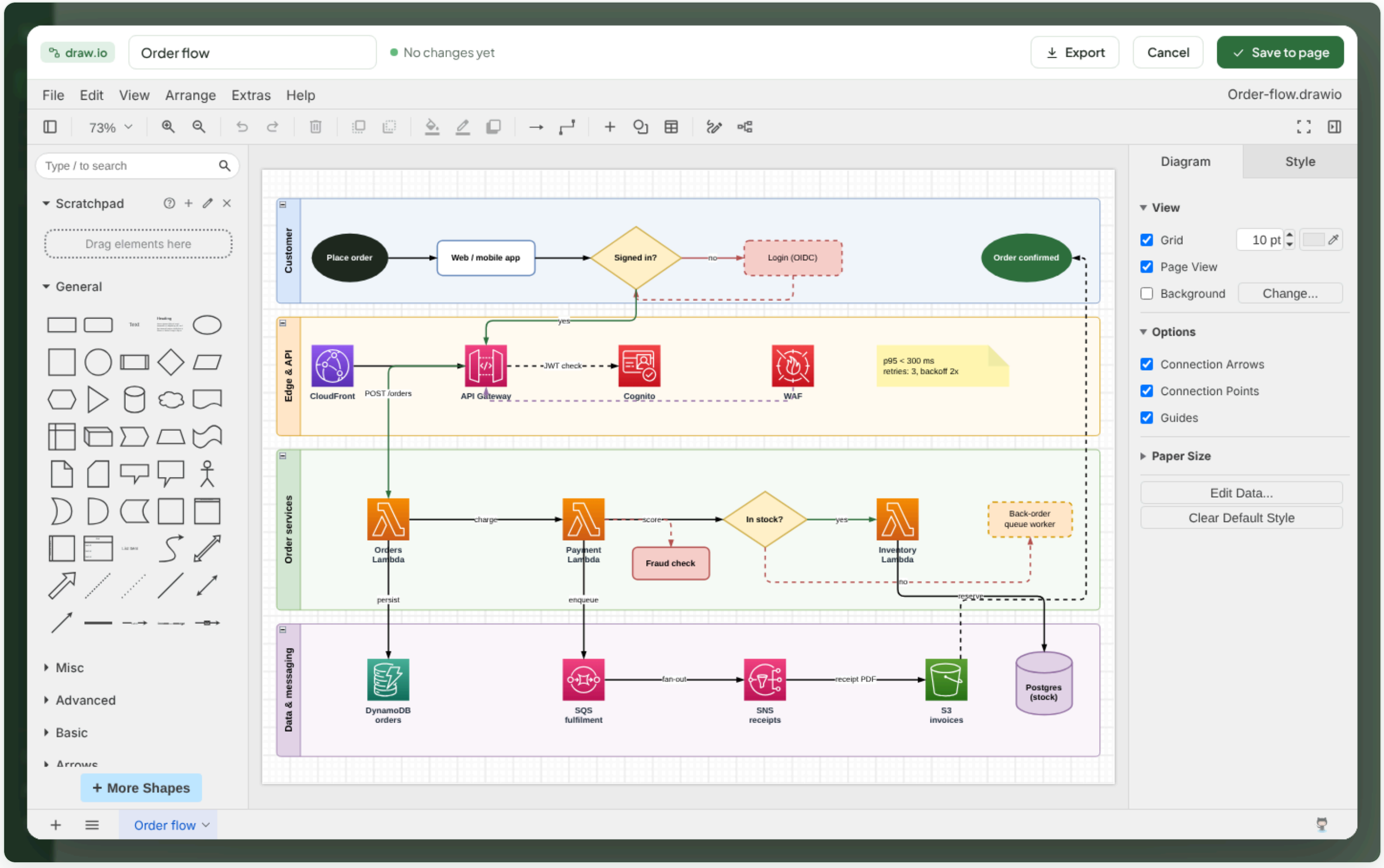

No browser available: plain preview

```
→ docs_write(action=diagram_create, project_id=2e333609-abff-4ce2-ba4e-6f8bb81d057e, title=Order flow (no browser), path=/tmp/claude-0/demo2/work/no-browser.drawio)
```


```
← {"id":"dg_3169bb20","label":"Order flow (no browser)","version":1,"markdown":"[Order flow (no browser)](diagram:dg_3169bb20)","preview":"plain: boxes, arrows and
text only; shapes beyond those show as boxes until someone opens the …
```

6. A person opens the agent's file in the real editor

Nothing was lost on the way: the file is the full draw.io drawing, shown here in the editor with the agent's Fraud check step.

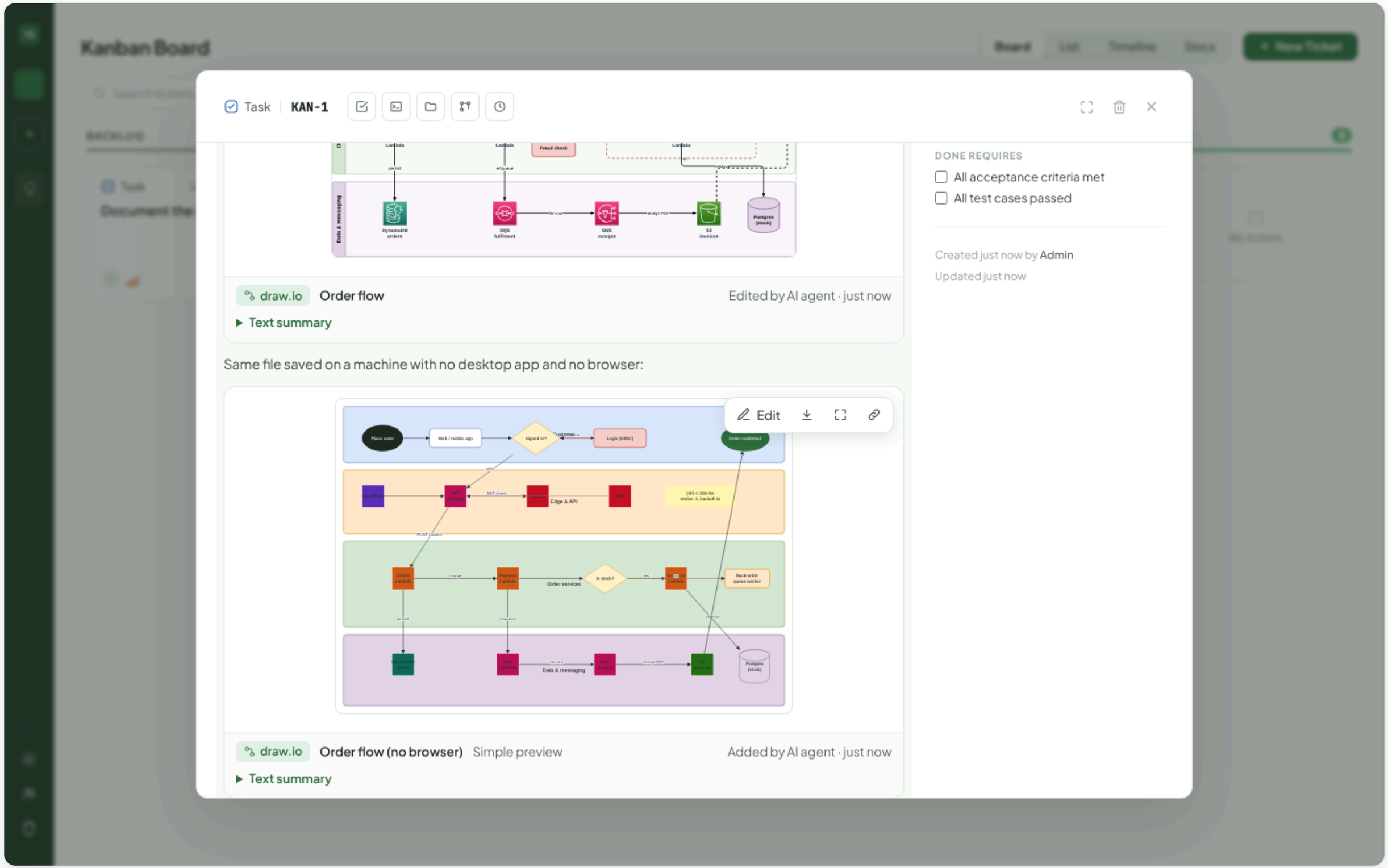


7. No desktop app, no browser: the plain preview, and the agent knows

The same file saved with the export switched off (like a bare server with no Chrome). The result says what was stored.

8. What people see in that case

Below the exact diagram: the same file saved without a browser. The footer says "Simple preview"; Edit opens the editor and saving replaces it with the full drawing.



What was run

Server	ruff check · ruff format --check · pytest: 454 passed, 1 skipped (a test that needs a real editor and browser; it passed here against draw.io's webapp and Chromium) — 13 new tests for export, privacy and preview kinds
Desktop	node --test: 34 passed (argument parsing, origin filter, export command, MCP entry env). The headless export was also run for real with Electron under Xvfb: 2–3 s, exit 0, identical SVG to Chrome's; a dead editor exits 1 in 0.5 s
Frontend	prettier · eslint (0 errors, 40 warnings = existing baseline) · tsc + build
Real run	MCP stdio client → server → desktop app's Chromium → real draw.io webapp; every picture in this document comes from that run
Migration	diagram_version.preview_kind (editor | agent | basic); upgrade verified on an empty database

Not covered: a packaged .dmg/.exe was not built here (the same code was run from source with Electron); macOS/Windows paths for finding Chrome/Edge are untested; a server on a headless Linux box without Chrome keeps the plain preview.